Project Design Phase – II

Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	4 Marks

Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored. Below are the DFD Level 0 and Level 1 for the Hyderabad Metro Rail Digital Ticketing System.

DFD Level 0 – Context Diagram (System Overview)

The Level 0 DFD shows the system as a single process and its interaction with external entities.

External Entity	Data Flow (Input to System)	Process	Data Flow (Output from System)	External Entity
Commuter (End User)	Booking Request (Source, Destination, Passengers, Journey Type, Payment Mode)	**Metro Digital Ticketing System (ServiceNow)**	QR Code Ticket + Booking Confirmation + Fare Amount	Commuter (End User)
Station Manager	Login Credentials + Access Request	**Metro Digital Ticketing System (ServiceNow)**	Booking Reports + Analytics Dashboard	Station Manager
IT Administrator	System Config + ACL Rules + Flow Activation	**Metro Digital Ticketing System (ServiceNow)**	System Status + Audit Logs	IT Administrator
Payment Gateway (Future)	Payment Confirmation + Transaction ID	**Metro Digital Ticketing System (ServiceNow)**	Payment Request + Amount	Payment Gateway (Future)

DFD Level 1 – Detailed Process Breakdown

The Level 1 DFD breaks down the core system process into sub-processes and data stores.

Process #	Process Name	Input Data	Output Data	Data Store
P1	User Authentication & Portal Access	Username + Password	Session Token / Portal Access	User Table (sys_user)
P2	Catalog Item Display & Selection	Portal Search Query	Book A Metro Ticket Form	Service Catalog (sc_cat_item)
P3	Form Variable Capture	Source, Destination, No. of Passengers, Journey Type, Payment Mode	Populated Variable Set	Catalog Variables (sc_item_option)
P4	Fare Auto-Calculation (Client Script)	Source sys_id + Destination sys_id + Journey Type + No. of Passengers	Fare Amount (Single / Return)	Metro Station Details Table

P5	Form Validation (Client Script)	No. of Passengers field input	Validation Alert / Pass	None (client-side)
P6	QR Code Generation (OnSubmit Script)	Submitted Form + Requested Item sys_id	QR Code URL via GlideModal	None (client-side display)
P7	Flow Designer Trigger (Metro Project)	Requested Item (RITM)	Flow Execution Signal	Flow Designer Engine
P8	Get Catalog Variables (Flow Action 1)	RITM sys_id	Variable Values (all fields)	sc_item_option table
P9	Create Record (Flow Action 2)	Mapped Variable Values	New Booking Record	Metro Database (custom table)
P10	ACL Enforcement	User Role + Table Access Request	Access Granted / Denied	ACL Rules (sys_security_acl)
P11	Email Notification	Booking Record + User Email	Booking Confirmation Email	Notification Engine

User Stories

User stories define the functionality of the system from the perspective of each user type. Each story follows the format: As a [user], I can [action] so that [benefit].

User Type	Epic (FR)	USN	User Story / Task	Acceptance Criteria	Priority	Release
Commuter (Portal User)	Ticket Booking	USN-001	As a commuter, I can open the ServiceNow po[illegible]	Catalog [illegible] portal.	High	Sprint 1
Commuter (Portal User)	Ticket Booking	USN-002	As a commuter, I can select my starting station [illegible]	[illegible] metro stations.	High	Sprint 1
Commuter (Portal User)	Ticket Booking	USN-003	As a commuter, I can enter the number of pas[illegible]	[illegible] values; alert shown for invalid input.	High	Sprint 1
Commuter (Portal User)	Fare Calculation	USN-004	As a commuter, I can see the fare auto-popula[illegible]	Correct fare amount [illegible] no manual entry ne	High	Sprint 1
Commuter (Portal User)	Journey Type	USN-005	As a commuter, I can select Single or Return jo[illegible]	Single [illegible] 2x base; Correct field populated.	High	Sprint 1
Commuter (Portal User)	Payment	USN-006	As a commuter, I can select my payment mode [illegible]	[illegible] Payment Mode' field appears	High	Sprint 1
Commuter (Portal User)	QR Code	USN-007	As a commuter, upon submitting the booking fo[illegible]	[illegible] with a scannable	High	Sprint 1
Commuter (Portal User)	Ticket Booking	USN-008	As a commuter, I can submit the form and get a [illegible]	RITM is generated [illegible] 'My Requests'.	High	Sprint 1
Commuter (Portal User)	Metro Card	USN-009	As a commuter, I can choose 'Recharge Metro [illegible]	[illegible] from same catalog.	Medium	Sprint 2
Commuter (Portal User)	Notifications	USN-010	As a commuter, I receive an email confirmation [illegible]	Email is sent [illegible] within 1 minute of submission.	High	Sprint 2
Station Manager	Data Access	USN-011	As a station manager, I can view all booking re[illegible]	[illegible] role; records show all mapped field	High	Sprint 1
Station Manager	Reporting	USN-012	As a station manager, I can filter bookings by d[illegible]	[illegible] on all key columns.	Medium	Sprint 2
Station Manager	Analytics	USN-013	As a station manager, I can view a dashboard o[illegible]	Dashboard widgets show [illegible] top routes, and peak time.	Medium	Sprint 3
IT Admin	Access Control	USN-014	As an IT admin, I can configure ACL rules so th[illegible]	[illegible] to Metro Database t	High	Sprint 1
IT Admin	Flow Config	USN-015	As an IT admin, I can activate and monitor the [illegible]	[illegible] after each submissio	High	Sprint 1
IT Admin	Maintenance	USN-016	As an IT admin, I can add or update metro stati[illegible]	New station is [illegible] within 5 minutes.	High	Sprint 1